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BIOTHON 2026

ORGANIZED BY DEPARTMENT OF BIOINFORMATICS AND SPONSORED BY ROYAL SOCIETY OF BIOLOGY AND IEEE EMBS

PROBLEM STATEMENTS

3 Domains | 20 Challenges | Building for Impact

DOMAINS

HEALTH CARE

AGRICULTURE

BIODIVERSITY & ENVIRONMENT

HEALTH CARE

Wearable AI Sensor for Fall Prevention in Elderly

Design a wearable device with AI algorithms that detects when a senior's centre of gravity shifts dangerously, providing real-time tactile/audio/visual alerts to prevent falls before they happen.

Wearable | **AI/ML** | **Elderly Care** | **IMU Sensors**

Why it fits: Great for hardware + ML teams. Falls cause 30% of disability in elderly compelling pitch. Can be prototyped with a smartphone as IMU substitute.

HEALTH CARE

Digital Mental Health and Psychological Support System for Students

Create a digital platform offering mental health resources, mood tracking, anonymous counselling connections, and AI-based early warning for students in higher education facing psychological distress.

Mental Health | **AI Screening** | **Students** | **Anonymous Platform**

Why it fits: Extremely relatable to college-age students. Combines app development with sentiment analysis and ethical AI design.

Non-Invasive Blood Vessel Blockage Detection (Alternative to Angiography)

Develop a cost-effective, non-invasive technology to detect blockages in blood vessels without radioactive contrast agents making vascular diagnostics safer and more accessible in rural clinics.

BioTech | **Medical Device** | **Non-invasive** | **Diagnostics**

Why it fits: High-impact, unique problem. Teams can propose ultrasound/near-IR optical sensing concepts with simulation data.

HEALTH CARE

Queuing and Bed Management System for Hospital OPDs

A smart hospital management solution integrating queuing models for OPDs, real-time bed availability, patient admission tracking, and medicine inventory reducing wait times and improving workflow.

Hospital Management **Real-time** **Dashboard** **Queue Optimisation**

Why it fits: Clear problem scope with well-defined use cases. Students can simulate OPD flow and show dashboards effectively.

AI-Driven Public Health Chatbot for Disease Awareness

Build a conversational AI chatbot that educates citizens about disease prevention, symptoms and nearest healthcare resources deployable in regional languages for rural populations.

NLP Chatbot **Multilingual** **Public Health**

Why it fits: LLM/RAG-based solution students can build quickly. Massive impact potential. Easy to demo interactively.

HEALTH CARE

Open Innovations

Have a groundbreaking idea in Health Care? Present your innovatition solution or prototype that addresses a critical challenge not covered by the other problem statements.

AGRICULTURE

Farmers Disease Diagnostic/Reporting Portal - AI Based Web App

An AI-powered mobile portal where farmers photograph disease symptoms on crops or livestock, receive AI diagnosis and suggested treatments, and alerts are sent to the nearest veterinarian or agri expert.

AI Diagnosis | **Image Recognition** | **Mobile** | **Alert System**

Why it fits: Combines NLP + CV + mobile development. Strong storytelling opportunity directly saves farmer livelihoods.

AGRICULTURE

Smart Irrigation System for Precision Farming

Design an IoT + sensor-based system to monitor real-time soil moisture and weather, then auto-schedule irrigation reducing water wastage and maximising crop yield through a mobile dashboard.

IoT | Water Conservation | Sensors | Dashboard

Why it fits: Excellent for hardware + software teams. Simulation mode possible without physical hardware for idea round.

Mobile App/ Web App for Direct Market Access for Farmers

Connect farmers directly with consumers and retailers through a marketplace app eliminating middlemen, enabling price negotiation, produce listing, and secure transaction management.

Marketplace | Agri Commerce | Mobile

Why it fits: Full-stack web/app challenge with clear user stories. Easily demonstrable with a working prototype.

AGRICULTURE

Smart Crop Advisory System for Small and Marginal Farmers

Create an advisory platform that recommends crop types, planting schedules and practices tailored to small farmers' land size, soil data and local weather bridging the knowledge gap in rural India.

Advisory System | Farmer App | Data Analytics

Why it fits: Broad enough for creative UI/UX solutions. Teams can integrate open soil and weather APIs without needing hardware.

AI-Driven Crop Disease Prediction and Management System

Develop an AI system that analyses crop images and environmental data to predict disease outbreaks early, giving farmers actionable treatment recommendations via a mobile/web app.

AI/ML | Computer Vision | Web App | Real-time Data

Why it fits: Visual dataset exists (crop images), well-scoped ML problem, high real-world impact. Students can demo with transfer learning on leaf images.

AGRICULTURE

Open Innovations

Have a groundbreaking idea in Agriculture? Present your innovatition solution or prototype that addresses a critical challenge not covered by the other problem statements.

ENVIRONMENT & BIODIVERSITY

AI-Driven Unified Data Platform for Oceanographic & Fisheries Insights

Build a unified AI platform integrating oceanographic datasets, fisheries records and molecular biodiversity data providing predictive insights on fish migration, marine health and sustainable fishing zones.

AI Platform **Big Data** **Geospatial ML** **Marine Science**

Why it fits: Ambitious, multi-data-source challenge ideal for final-year teams. Open APIs from INCOIS available. Strong differentiation from typical hackathon entries.

ENVIRONMENT & BIODIVERSITY

Identifying Taxonomy and Assessing Biodiversity from eDNA Datasets

Use environmental DNA (eDNA) sequencing data to identify species taxonomy and assess biodiversity in aquatic ecosystems a bioinformatics pipeline combining sequence alignment, ML classification and ecological reporting.

Bioinformatics | **eDNA** | **ML Classification** | **Biodiversity**

Why it fits: Rare bioinformatics problem in national hackathons. Great for Biology + CS cross-disciplinary teams. Public eDNA datasets available on NCBI/EBI.

AI/ML Models for Predicting Prices of Agricultural Commodities

Develop predictive models for forecasting prices of pulses and vegetables (onion, potato, tomato) using historical trends, seasonality, market intelligence and satellite crop data powering smarter government interventions.

Time-series ML | **ARIMA** | **Price Prediction** | **Data Science**

Why it fits: Well-defined ML regression problem with open government datasets. Students can compare ARIMA vs LSTM and present clear metrics.

ENVIRONMENT & BIODIVERSITY

FloatChat - AI Conversational Interface for Ocean Data Discovery

Create a natural language chatbot interface that lets researchers and citizens query and visualise oceanographic data from ARGO floats making complex scientific datasets accessible through plain-English conversation.

LLM | Data Visualisation | Science Portal

Why it fits: Excellent LLM + API integration challenge. Public ARGO data is freely accessible. Great for teams interested in conversational AI.

AI/ML Solution for Detecting Face-Swap Deepfake Videos

Build a deep learning model that detects face-swap deepfake videos by analysing spatial, temporal and frequency anomalies outputting a detailed forensic report on whether a video is authentic or manipulated.

Deep Learning | CNN/RNN | Computer Vision | Cybersecurity

Why it fits: Hot research area with open datasets (FaceForensics++, Celeb-DF). Students can fine-tune pre-trained models for quick results.

ENVIRONMENT & BIODIVERSITY

Real-time Ganga River Water Quality Forecasting using AI and IoT

Build an AI-enabled Decision Support System integrating satellite data, IoT sensors and hydrological models to forecast Ganga water quality (DO, BOD, pathogens) 3–5 days ahead with public mobile dashboard and alerts.

Satellite Data | **IoT** | **Water Quality** | **AI/DSS**

Why it fits: Multi-data-source, high-visibility national challenge. Open datasets available via CPCB and IMD. Powerful demo potential with map-based visualisation.

ENVIRONMENT & BIODIVERSITY

Industrial-Scale Green Hydrogen Production Facility Concept

Design a detailed concept for a green hydrogen facility producing 50+ tonnes/day at under $2/kg levelised cost including electrolyser sizing, renewable energy model, CAPEX/OPEX breakdown and carbon credit estimation.

Green Energy | **Techno-Economics** | **Hydrogen** | **Modelling**

Why it fits: Ideal for engineering and management students.
No hardware needed pure analysis and modelling. Strong presentation component.

Blockchain-Based Blue Carbon Registry and MRV System

Create a blockchain-based registry to measure, report and verify (MRV) blue carbon stored in coastal ecosystems (mangroves, seagrass) enabling transparent carbon credit issuance and climate finance tracking.

Blockchain | **Carbon Credits** | **Climate Tech** | **Coastal Ecology**

Why it fits: Combines Web3 + climate science. Strong differentiation. Immutable registry concept is achievable with Ethereum/Polygon testnet. Great for CS + Env. Science teams.

ENVIRONMENT & BIODIVERSITY

Digital Water Footprint Calculator for Agricultural Products

Develop a user-friendly web/mobile app where users scan or enter food items to instantly learn their water footprint driving awareness about virtual water consumption in agriculture and promoting conservation behaviour.

Sustainability App | AI Scan | Water Conservation | Education

Why it fits: Simple concept with broad public appeal. Camera + image recognition + database lookup. Ideal for early-stage teams needing a clean, demoable product.

Waste to Energy - Innovative MSW Recycling and Community Solution

Design a community-based digital tracking and management system for municipal solid waste from segregation at source through collection, recycling and conversion to bioenergy, with landfill gas monitoring.

Waste Management | IoT Tracking | Dashboard

Why it fits: Clear problem scope, highly visible real-world challenge. Teams can build a full digital tracking MVP without hardware.

ENVIRONMENT & BIODIVERSITY

Open Innovations

Have a groundbreaking idea in Environment & Biodiversity? Present your innovatition solution or prototype that addresses a critical challenge not covered by the other problem statements.

🏆 REWARDS

Prizes Per Domain

🥇 1st Prize – ₹30,000 Cash
🥈 2nd Prize – ₹20,000 Cash
🥉 3rd Prize – ₹10,000 Cash

THANK YOU

3 Domains | 20 Challenges | Building for Impact

SOLVE REAL WORLD PROBLEMS

AND WIN CASH PRIZES